

7 C apply knowledge from Newton's cradle that transfer of KE between 2 masses is 100% if masses are equal and one mass is initially stationary

we model the system as separated blocks doing elastic collisions, take right +ve:



consider A and B only:

$$\text{PCLM: } m_A u_A + 0 = m_A v_A + m_B v_B$$

$$\text{relative speed: } v_B - v_A = 10$$

$$m(10) = m v_A + (4m)(v_A + 10)$$

$$10 = 5v_A + 40$$

$$v_A = -\frac{30}{5} = -6 \text{ m s}^{-1}$$

$$v_B = v_A + 10 = 4 \text{ m s}^{-1}$$

so A will move left. B will undergo newton's cradle style collision until E moves to hit F

consider E and F only

$$\text{PCLM: } m_E u_E + 0 = m_E v_E + m_F v_F$$

$$\text{relative speed: } v_F - v_E = 4$$

$$(4m)(4) = (4m)v_E + m(4 + v_E)$$

$$16 = 5v_E + 4$$

$$v_E = \frac{12}{5} = +2.4 \text{ m s}^{-1}$$

$$v_F = \frac{32}{5} = +6.4 \text{ m s}^{-1}$$

so both E and F moves to the right. that leaves B, C and D at rest.

**8** **B** sketch in order visualize

**3**